

- `approve(spender, amount)` – allows “spender” to spend “amount” tokens on behalf of the account owner
- `allowance(owner address, spender address)` – returns the number of tokens the “spender address” can spend on behalf of the “owner address”

The contract will reject transfers involving insufficient balances or unauthorized spending. The first four functions – reading supply, balances, and sending tokens – are intuitive and expected. The last two functions – `approve` and `allowance` – are critical to understanding the power of the ERC-20 interface. Without these functions, users would be limited to directly transferring tokens to and from accounts. With approval functionality, contracts (or trusted accounts) can be whitelisted to act as custodians for a user’s tokens without directly holding the token balance. This widens the scope of possible applications because users retain full custody before an approved spender executes a transaction.

There are three main ERC-20 token main categories, but tokens can be in more than one at the same time.

Equity Token

An equity token – not to be confused with equities or stocks in the traditional finance sense – represents ownership of an underlying asset or pool of assets. The units must be fungible so that each corresponds to an identical share in the pool. For example, suppose a token TKN has a total fixed